CHINTAN VAGHAMSHI

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EDUCATION

Indian Institute of Technology Jodhpur — Aug 2024 – Jul 2028
B.Tech in Electrical Engineering | CGPA: 7.50/10.00 — *Jodhpur, Rajasthan*

Ashadeep Vidhyalaya — May 2022 – May 2024
Intermediate | Percentage : 87% — *Surat , Guajarat*

Ashadeep Vidhyalaya — May 2020 – May 2022
Matriculation | Percentage : 95% — *Surat , Guajarat*

TECHNICAL SKILLS

Languages: C, C++, Python, JavaScript , HTML, CSS

Technologies: Node.js, React, Express, MongoDB, Redis, BullMQ, Socket.io, Microservices, REST APIs, WebSockets, JWT, WebCrypto API

Tools: Git, Vercel, Render, Cloudflare R2, Cloudinary, Postman, VS Code

EXPERIENCE

Open Source Contributor | NetworkX — Nov 2025 – Dec 2026
Python, Graph Theory, Open Source — *Remote*

– Analyzed the NetworkX graph-processing codebase, focusing on algorithm correctness across graph traversal APIs.
– Enhanced documentation for core graph APIs by developing a new reference page for reporting views, integrated into the main documentation site; reduced onboarding time for new contributors by 30%.

PROJECTS

Lithium-matcher | Github | Video Explanation | C++20, Python (Synthetic Data Generation), GCC / MinGW, CMake

– Engineered a C++20 matching engine achieving a peak threaded throughput of **17.2M orders/sec**, with a core **p50 latency of 28ns** and **p99 of 150ns** (verified via $-O3$ optimized hardware cycle counter benchmarking).
– Developed a 240MB fixed-capacity Memory Pool and an **O(1) Limit Order Book**, yielding a **32x allocation speedup** and guaranteeing zero-copy, in-place updates.
– Implemented a lock-free SPSC ring buffer to decouple ingestion from matching; resolved a critical wraparound aliasing bug in buffer-full detection that caused silent order drops.
– Devised the Limit Order Book using **std::array** indexed directly by price levels to achieve strict **O(1) time complexity** for all order insertions and matches, completely eliminating the **O(log N)** traversal overhead typical of **std::map** implementations.
– Profiled the full pipeline in isolated stages across 3M Python-generated synthetic orders, separating matching engine cost from I/O overhead to avoid conflated measurements and isolate true matching engine performance.

SecureShare | Github | Live | Video Explanation | JavaScript, React, Node.js, Express, MongoDB, Cloudinary, WebCrypto API

– Designed a **zero-knowledge E2E encrypted** file sharing platform across **18 REST APIs** — browser encrypts files via **WebCrypto (AES-256-GCM + RSA-2048 OAEP)** before upload; **server mathematically incapable of reading user files** even under full infrastructure compromise.
– Implemented **PBKDF2 key derivation** (100K iterations), MongoDB-persisted account lockout, **JWT refresh token rotation** with HttpOnly cookies, and **SHA-256 integrity verification** — defeating **4 simultaneous attack vectors** (XSS, CSRF, brute force, tampering).
– Established a complete security audit trail across upload, download, share, revoke, and auth events, with a user dashboard surfacing real-time failed login attempts and **INTEGRITY_FAIL** alerts — **zero plaintext file exposure** verified across MongoDB, Cloudinary, and server RAM.

WayPoint | Github | Live | JavaScript, React, Node.js, Express, MongoDB, Redis, BullMQ, Socket.io, Cloudflare R2

– Spearheaded a real-time collaborative platform (MERN) with Socket.io.
– Refactored task-update endpoint to use a non-blocking 'Fire-and-Forget' background write pattern, reducing server Time To First Byte (TTFB) from **2.33s to 1.10s (53% reduction)**.
– Engineered JWT auth with access/refresh token rotation and 4-tier RBAC (owner/admin/member/viewer) across **36 REST endpoints**; integrated collaborative document editing using **Y.js (CRDTs)** for conflict-free multi-user sync.
– Designed Redis (Upstash) caching layer with event-driven invalidation (**85% hit rate**); tuned BullMQ worker polling (1s→30s), cutting idle DB requests **30x (170k→5.7k/day)** to drastically reduce infrastructure costs.

COURSEWORK

- Data Structures & Algorithms
- Graph Theory
- Pattern Recognition & Machine Learning
- Digital Design
- Probability & Statistics
- Artificial Intelligence

ACHIEVEMENTS & PARTICIPATION

– Solved **300+** DSA problems across LeetCode and competitive programming platforms.
– Achieved **Gold Medal** In SOF Science Olympiad
– Participated in **Smart India Hackathon** (national-level government hackathon).
– Ranked **3rd in State** in Karate; competed at **National Level twice**.

LEADERSHIP

– **Core Member**, E-Cell, IIT Jodhpur — organized and managed entrepreneurship events over 9 months.
– **Core Member**, Programming Club, IIT Jodhpur — co-organized competitive programming contests and workshops.
– **Core Member**, SWC IIT Jodhpur, , Mentored 10 freshers academically and non-academically.